EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

Lattice and Lattice-Ladder filter structures are among the most robust digital filter topologies. Widely used in speech processing (Linear Predictive Coding - LPC) and adaptive filtering, lattice structures parameterized by reflection coefficients k_m (PARCOR coefficients) guarantee stability simply by checking $|k_m| < 1$. This lecture also surveys Finite Word-Length Effects (FWE).

Pedagogical Strategy:

1. Derive the recursive FIR Lattice equations:
$$f_m[n] = f_{m-1}[n] + k_m g_{m-1}[n-1]$$
$$g_m[n] = k_m f_{m-1}[n] + g_{m-1}[n-1]$$
2. Derive the step-down algorithm to convert direct-form polynomials $A_m(z)$ to reflection coefficients k_m.
3. Formulate the All-Pole IIR Lattice and Pole-Zero Lattice-Ladder structure.
4. Prove the **Schur-Cohn Stability Criterion**: An all-pole filter is stable if and only if $|k_m| < 1$ for all $m = 1, 2, \dots, N$.
5. Survey Finite Word-Length Effects: Coefficient quantization, Round-off noise variance $\sigma_e^2 = \frac{2^{-2B}}{12}$, and zero-input limit cycles.

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Convert** FIR direct form coefficients to lattice reflection coefficients k_m using the step-down algorithm.
2. **Evaluate** IIR filter stability directly from reflection coefficients.
3. **Construct** signal flow graphs for FIR lattice and IIR lattice-ladder filters.
4. **Quantify** round-off noise power and identify limit cycle conditions.

2. MATHEMATICAL FOUNDATIONS

2.1 FIR Lattice Recursion & Step-Down Algorithm

- **Lattice Recursion:**

$$A_m(z) = A_{m-1}(z) + k_m z^{-1} B_{m-1}(z)$$
$$B_m(z) = z^{-m} A_m(z^{-1}) \quad \text{(Reverse Polynomial)}$$

- **Step-Down Algorithm (Direct $\to$ Lattice):** Given $A_N(z) = 1 + \sum_{i=1}^{N} \alpha_N(i) z^{-i}$:
    1. Set $k_N = \alpha_N(N)$.
    2. For $m = N, N-1, \dots, 2$:

$$\alpha_{m-1}(i) = \frac{\alpha_m(i) - k_m \alpha_m(m-i)}{1 - k_m^2}, \quad i = 1, 2, \dots, m-1$$
$$k_{m-1} = \alpha_{m-1}(m-1)$$

2.3 Finite Word-Length Effects Summary

1. **Coefficient Quantization:** Shifts poles in the z-plane; high-order direct forms are severely sensitive; lattice and cascade forms are highly robust.
2. **Round-off Noise:** Fixed-point truncation error modeled as white noise with variance $\sigma_e^2 = \frac{2^{-2B}}{12}$ for B-bit quantization. Output noise power:
$$\sigma_y^2 = \sigma_e^2 \sum_{n=0}^{\infty} |h[n]|^2 = \sigma_e^2 \frac{1}{2\pi j} \oint H(z) H(z^{-1}) z^{-1} dz$$
3. **Limit Cycle Oscillations:** Nonlinear oscillations in recursive filters caused by arithmetic quantization (deadbands) and two's complement overflow. Mitigated by using saturation arithmetic.

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3. WORKED NUMERICAL EXAMPLES

Example 20.1: Step-Down Conversion to Reflection Coefficients

Problem: Find the reflection coefficients k_1, k_2, k_3 for the FIR filter:
$$A_3(z) = 1 - 0.9z^{-1} + 0.64z^{-2} - 0.5z^{-3}$$

Solution:

- **Stage 3:** $k_3 = \alpha_3(3) = -0.5.\ 1 - k_3^2 = 1 - (-0.5)^2 = 1 - 0.25 = 0.75.$
- **Compute $\alpha_2(i)$ for $m = 3$:**
    - $\alpha_2(1) = \frac{\alpha_3(1) - k_3\alpha_3(2)}{0.75} = \frac{-0.9-(-0.5)(0.64)}{0.75} = \frac{-0.9+0.32}{0.75} = \frac{-0.58}{0.75} = -0.7733.$
    - $\alpha_2(2) = \frac{\alpha_3(2) - k_3\alpha_3(1)}{0.75} = \frac{0.64-(-0.5)(-0.9)}{0.75} = \frac{0.64-0.45}{0.75} = \frac{0.19}{0.75} = 0.2533.\ k_2 = \alpha_2(2) = 0.2533.$
- **Stage 2:** $1 - k_2^2 = 1 - (0.2533)^2 = 1 - 0.0642 = 0.9358.$
- **Compute $\alpha_1(i)$ for $m = 2$:**
    - $\alpha_1(1) = \frac{\alpha_2(1) - k_2\alpha_2(1)}{0.9358} = \frac{-0.7733(1-0.2533)}{0.9358} = \frac{-0.7733\times 0.7467}{0.9358} = -0.6170.\ k_1 = \alpha_1(1) = -0.6170.$
- **Stability Check:** $|k_1| = 0.6170 < 1, ; |k_2| = 0.2533 < 1, ; |k_3| = 0.5000 < 1$. All $|k_m| < 1 \implies$ The corresponding all-pole filter is **Stable**.

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) State the Schur-Cohn stability criterion for digital filters. Using the step-down algorithm, determine if the system is stable:

$$H(z) = \frac{1}{1 - 1.2z^{-1} + 0.8z^{-2} - 0.2z^{-3}}$$

(9 Marks) **(b)** Discuss limit cycle oscillations in recursive digital filters. Explain the difference between zero-input granular limit cycles and overflow oscillations. *(6 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Statement of Schur-Cohn criterion: All $|k_m| < 1$ *(2 Marks)*
  - Step-down execution: $k_3 = -0.2 \implies 1 - k_3^2 = 0.96$
    $\alpha_2(1) = \frac{-1.2-(-0.2)(0.8)}{0.96} = \frac{-1.04}{0.96} = -1.0833$
    $\alpha_2(2) = \frac{0.8-(-0.2)(-1.2)}{0.96} = \frac{0.56}{0.96} = 0.5833 \implies k_2 = 0.5833\ 1 - k_2^2 = 1 - 0.3403 = 0.6597$
    $\alpha_1(1) = \frac{-1.0833-(0.5833)(-1.0833)}{0.6597} = \frac{-1.0833(0.4167)}{0.6597} = -0.6843 \implies k_1 = -0.6843$ *(5 Marks)*
  - Since $|k_1| = 0.6843 < 1, |k_2| = 0.5833 < 1, |k_3| = 0.2 < 1 \implies$ **Stable** *(2 Marks)*
- **Part (b):**
  - Explanation of granular limit cycles due to quantization rounding *(3 Marks)*
  - Explanation of overflow oscillations and mitigation via saturation arithmetic *(3 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# Verify step-down conversion
a = np.array([1.0, -1.2, 0.8, -0.2])
roots = np.roots(a)
print("Pole Magnitudes:", np.abs(roots))
print("All inside unit circle:", np.all(np.abs(roots) < 1.0))
```